

NAGHUL ADHITHYA VENKATESWARAN

✉ nav5@illinois.edu  [linkedin.com/in/naghuladhithya](https://www.linkedin.com/in/naghuladhithya)  github.com/naghuladhithya

EDUCATION

University of Illinois Urbana-Champaign

Bachelor of Science (B.S.) in Data Science & Information Sciences | Minor: Computer Science

Expected: May 2029

Champaign, IL

EXPERIENCE

The Ohio State University

Software Engineering Research Intern

May 2025 – Sept 2025

Columbus, OH

- Engineered benchmarking pipelines evaluating **human vs LLM-generated** object-oriented systems across programming workflows
- Analyzed abstraction quality, **modularity**, and reasoning efficiency in collaborative human-AI software engineering paradigms
- Developed reproducible evaluation frameworks for cognitive variation analysis in generative programming environments

Delhi Technological University

Machine Learning Systems Intern

May 2025 – July 2025

Remote

- Designed multimodal **biometric** authentication pipelines integrating keystroke dynamics, achieving **99.47%** classification accuracy
- Optimized deep learning feature extraction using **metaheuristic algorithms**, improving cross-modal verification scalability
- Evaluated distributed identity inference workflows under heterogeneous conditions for reliable real-time authentication systems

Indian Institute of Information Technology Tiruchirappalli

Generative AI Engineering Intern

May 2024 – July 2024

Remote

- Fine-tuned **FLUX.1-dev** diffusion architectures with **LoRA** adapters for controllable **sketch-to-image** generation rendering
- Optimized latent diffusion and prompt-conditioning workflows, improving texture preservation and generative consistency by **20%**
- Engineered **style-transfer** pipelines enabling scalable prompt-guided **colorization** and artistic image synthesis

Incognito Blueprints

ML & Data Analytics Research Intern

May 2023 – Aug 2023

San Francisco, CA

- Designed **ML pipelines** with ModelOps principles, boosting throughput **25%** for 20+ SMB clients using MySQL
- Optimized **decision systems** with real-time inference & performance analysis, improving accuracy by **20%** & reducing lag **30%**
- Published reproducible research enabling scalable experimentation; integrated **NLP** workflows into distributed ML systems

PROJECTS AND RESEARCH

Real-Time Driver Drowsiness Detection for Safe Driving | TensorFlow, OpenCV, Deep Learning

- Developed deep learning-based driver monitoring systems using Eye Aspect Ratio (EAR), facial landmarks, and HSV-based sunglass detection pipelines
- Engineered robust real-time inference workflows for safety-critical transportation monitoring environments
- Presented at **ICMEET 2024** and later published in **Springer** proceedings

Real-Time Vehicle Detection and Counting System | YOLOv8, Computer Vision

- Designed **YOLOv8**-based traffic analytics pipelines for real-time congestion monitoring and intelligent transportation analysis
- Implemented centroid-based multi-object tracking systems for scalable vehicle detection workflows
- Presented at **ICDSA 2024** and later published in **Springer** proceedings

Ethical AI Vision System | OpenCV, TensorFlow

- Led a four-member global team developing AI-powered driver monitoring systems achieving **95%** detection accuracy
- Integrated **OpenCV** and **TensorFlow** pipelines for real-time eye-tracking and drowsiness inference workflows
- Evaluated ethical AI deployment constraints including fairness, bias mitigation, and trustworthy decision-making systems

TECHNICAL SKILLS

Languages: C++, Python, JavaScript, SQL, HTML/CSS

Frameworks & Libraries: PyTorch, TensorFlow, Scikit-Learn, React, Flask, Node.js

Tools & Technologies: Git, MySQL, VS Code, Jupyter Notebook

Core Areas: Machine Learning, Deep Learning, Computer Vision, Generative AI, NLP